

VOLUME 5 • B.TECH CSE STUDY SERIES

CSS & CSS3

Complete Professional Handwritten Notes

*Box Model, Specificity & Cascade, Flexbox & Grid Layouts, Responsive
Media Queries, Animations & CSS Variables*

```
.container {  
  display: flex;  
  justify-content: center;  
}
```

✓ Box Model & Box Sizing
Mechanics

✓ Flexbox Complete Architecture

✓ CSS Grid 2D Layout System

✓ Specificity Calculation Rules

✓ Responsive Design & Breakpoints

✓ 30+ Viva & UI Interview Q&A

Designed for University Exams, UI/UX Engineering & Modern Web Development



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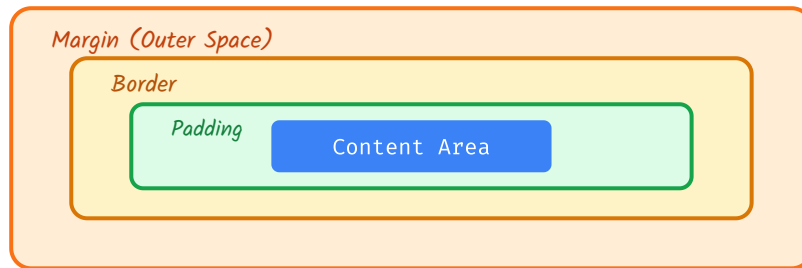
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CHAPTER 5: THE CSS BOX MODEL



CSS Box Model Anatomy

SVG DIAGRAM: CSS BOX MODEL ARCHITECTURE



Pro Tip: Box-Sizing

Always set `* { box-sizing: border-box; }` so that padding and border are included inside the specified width and height instead of expanding the element size!

```
/* Standard Box Model Reset */
* {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

.card {
  width: 300px;
  padding: 20px;
  border: 2px solid #0284c7;
  margin: 15px auto;
}
```

CHAPTER 7: FLEXBOX VS CSS GRID



Flexbox (1D) vs CSS Grid (2D) Comparison

Feature	Flexbox (Flexible Box)	CSS Grid
Dimensions	1-Dimensional (Row OR Column at a time).	2-Dimensional (Rows AND Columns simultaneously).
Primary Focus	Content-driven alignment (component level).	Layout-driven structure (page level layout).
Key Container Props	display: flex; justify-content; align-items;	display: grid; grid-template-columns; gap;

CHAPTER 14: 30+ VIVA & INTERVIEW QUESTIONS

Q1. What is CSS Specificity and how is it calculated?

Ans: Specificity decides which CSS rule applies when multiple rules target the same element. Calculated as: Inline Style (1000) > ID Selector (100) > Class/Attribute/Pseudo-class (10) > Element/Pseudo-element (1).

Q2. What is the difference between position: absolute and position: relative?

Ans: relative positions an element relative to its normal document flow. absolute removes element from document flow and positions it relative to its nearest ancestor with a non-static position.